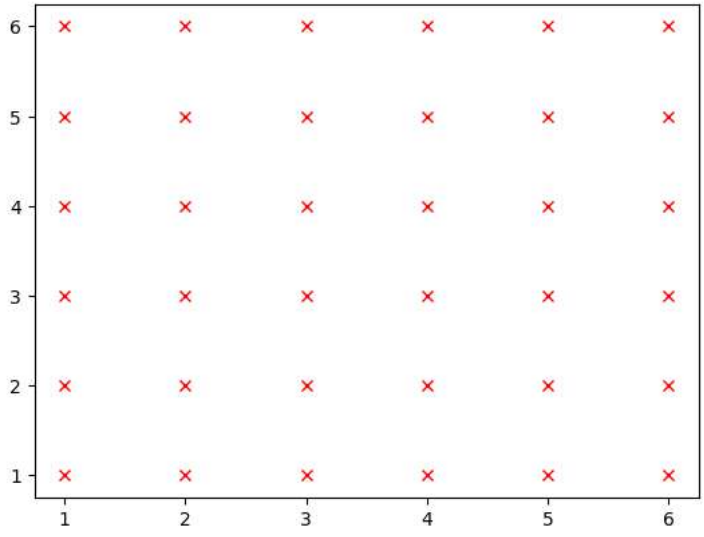
```
import numpy as np
import matplotlib.pyplot as plt

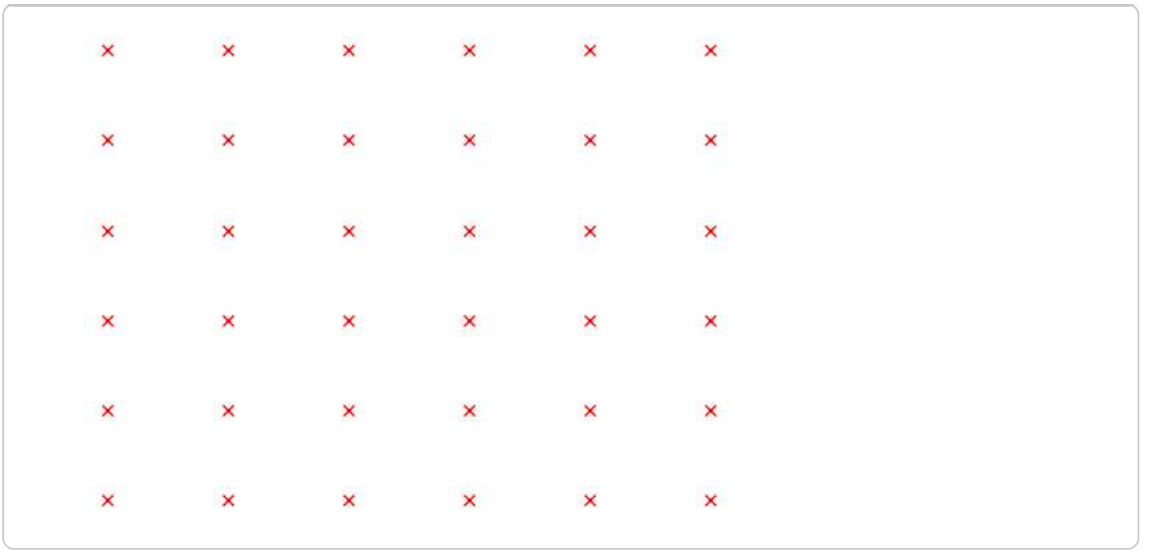
x=y= np.linspace(1, 6, 6)
X, Y = np.meshgrid(x, y)
fig,ax=plt.subplots()
plt.plot(X,Y,marker='x',color='red',ls='none')
print(X)
print(Y)
plt.show()
```

```
[[1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]]
[[1. 1. 1. 1. 1. 1.]
 [2. 2. 2. 2. 2. 2.]
 [3. 3. 3. 3. 3. 3.]
 [4. 4. 4. 4. 4. 4.]
 [5. 5. 5. 5. 5. 5.]
 [6. 6. 6. 6. 6. 6.]]
```




```
import numpy as np
import matplotlib.pyplot as plt

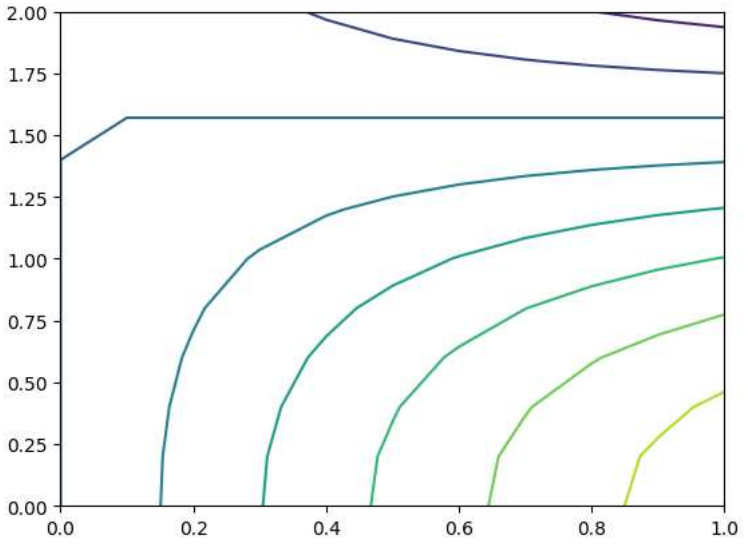
x=y= np.linspace(1, 6, 6)
X, Y = np.meshgrid(x, y)
fig,ax=plt.subplots()
plt.plot(X,Y,marker='x',color='red',ls='none')
ax.set_xticks([])
ax.set_yticks([])
ax.set_frame_on(False)
```



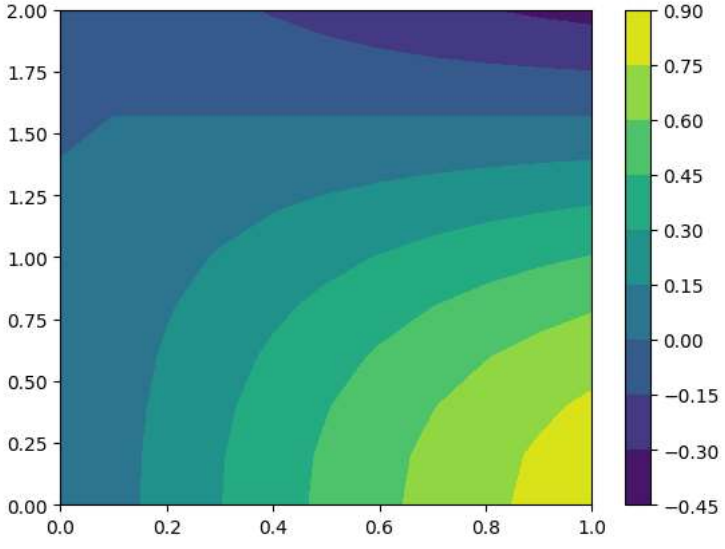
```
import numpy as np
import matplotlib.pyplot as plt
a=1
b=2
N=10
x= np.linspace(0, a, N+1)
y=np.linspace(0,b,N+1)
print(x)
print(y)
X, Y = np.meshgrid(x, y)
print(X)
print(Y)
```

```
[0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
[0.  0.2 0.4 0.6 0.8 1.  1.2 1.4 1.6 1.8 2. ]
[[0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
 [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]]
[[0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0. ]
 [0.2 0.2 0.2 0.2 0.2 0.2 0.2 0.2 0.2 0.2 0.2]
 [0.4 0.4 0.4 0.4 0.4 0.4 0.4 0.4 0.4 0.4 0.4]
 [0.6 0.6 0.6 0.6 0.6 0.6 0.6 0.6 0.6 0.6 0.6]
 [0.8 0.8 0.8 0.8 0.8 0.8 0.8 0.8 0.8 0.8 0.8]
 [1.  1.  1.  1.  1.  1.  1.  1.  1.  1.  1. ]
 [1.2 1.2 1.2 1.2 1.2 1.2 1.2 1.2 1.2 1.2 1.2]
 [1.4 1.4 1.4 1.4 1.4 1.4 1.4 1.4 1.4 1.4 1.4]
 [1.6 1.6 1.6 1.6 1.6 1.6 1.6 1.6 1.6 1.6 1.6]
 [1.8 1.8 1.8 1.8 1.8 1.8 1.8 1.8 1.8 1.8 1.8]
 [2.  2.  2.  2.  2.  2.  2.  2.  2.  2.  2. ]]
```

```
import numpy as np
import matplotlib.pyplot as plt
a=1
b=2
N=10
x= np.linspace(0, a, N+1)
y=np.linspace(0,b,N+1)
X, Y = np.meshgrid(x, y)
f=np.sin(X)*np.cos(Y)
plt.contour(X,Y,f)
plt.show()
```




```
import numpy as np
import matplotlib.pyplot as plt
a=1
b=2
N=10
x= np.linspace(0, a, N+1)
y=np.linspace(0,b,N+1)
X, Y = np.meshgrid(x, y)
f=np.sin(X)*np.cos(Y)
plt.contourf(X,Y,f)
plt.colorbar()
plt.show()
```



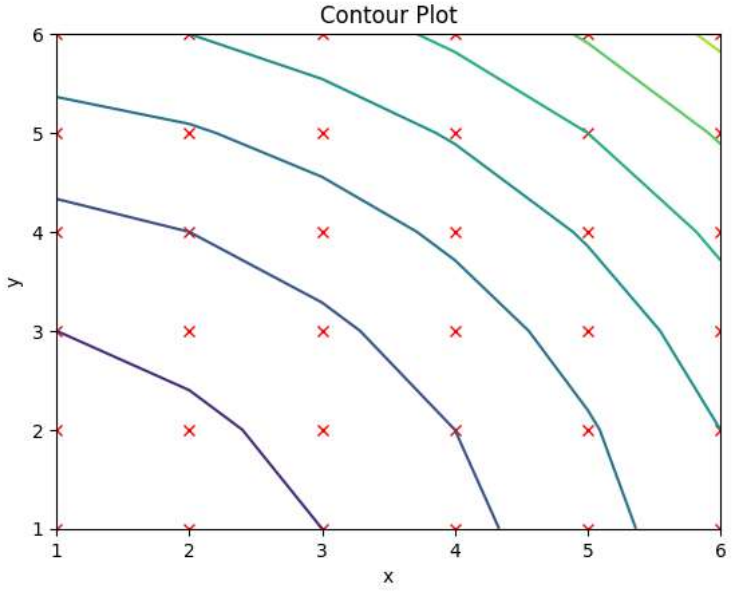

```
import numpy as np
import matplotlib.pyplot as plt

x=y= np.linspace(1, 6, 6)
X, Y = np.meshgrid(x, y)
fig,ax=plt.subplots()
plt.plot(X,Y,marker='x',color='red',ls='none')
Z = X**2 + Y**2
plt.contour(X, Y, Z)

plt.xlabel('x')
plt.ylabel('y')
```

```
plt.title('Contour Plot')

plt.show()
print(X)
print(Y)
print(Z)
plt.show()
```




```
[[1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]
 [1. 2. 3. 4. 5. 6.]]
[[1. 1. 1. 1. 1. 1.]
 [2. 2. 2. 2. 2. 2.]
 [3. 3. 3. 3. 3. 3.]
 [4. 4. 4. 4. 4. 4.]
 [5. 5. 5. 5. 5. 5.]
 [6. 6. 6. 6. 6. 6.]]
[[ 2.  5. 10. 17. 26. 37.]
 [ 5.  8. 13. 20. 29. 40.]
 [10. 13. 18. 25. 34. 45.]
 [17. 20. 25. 32. 41. 52.]
 [26. 29. 34. 41. 50. 61.]
 [37. 40. 45. 52. 61. 72.]]
```

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-3, 3, 100)
y = np.linspace(-3, 3, 100)

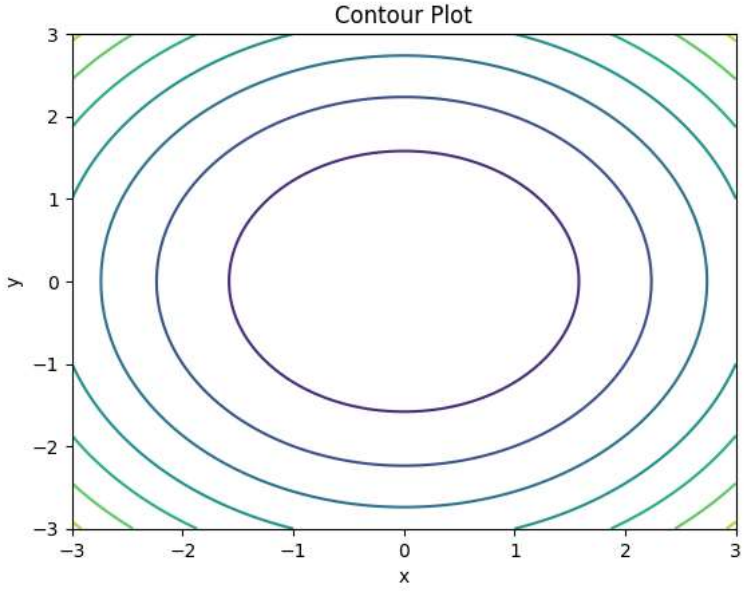
X, Y = np.meshgrid(x, y)

Z = X**2 + Y**2
plt.contour(X, Y, Z)

plt.xlabel('x')
plt.ylabel('y')
plt.title('Contour Plot')

plt.show()
```

```
[[-3.         -2.93939394 -2.87878788 ...  2.87878788  2.93939394
   3.        ]
 [-3.         -2.93939394 -2.87878788 ...  2.87878788  2.93939394
   3.        ]
 [-3.         -2.93939394 -2.87878788 ...  2.87878788  2.93939394
   3.        ]
 ...
 [-3.         -2.93939394 -2.87878788 ...  2.87878788  2.93939394
   3.        ]
 [-3.         -2.93939394 -2.87878788 ...  2.87878788  2.93939394
   3.        ]
 [-3.         -2.93939394 -2.87878788 ...  2.87878788  2.93939394
   3.        ]]
[[-3.         -3.         -3.         ... -3.         -3.
  -3.        ]
 [-2.93939394 -2.93939394 -2.93939394 ... -2.93939394 -2.93939394
  -2.93939394]
 [-2.87878788 -2.87878788 -2.87878788 ... -2.87878788 -2.87878788
  -2.87878788]
 ...
 [ 2.87878788  2.87878788  2.87878788 ...  2.87878788  2.87878788
   2.87878788]
 [ 2.93939394  2.93939394  2.93939394 ...  2.93939394  2.93939394
   2.93939394]
 [ 3.          3.          3.         ...  3.          3.
   3.        ]]
[[18.         17.64003673 17.28741965 ... 17.28741965 17.64003673
  18.        ]
 [17.64003673 17.28007346 16.92745638 ... 16.92745638 17.28007346
  17.64003673]
 [17.28741965 16.92745638 16.5748393  ... 16.5748393  16.92745638
  17.28741965]
 ...
 [17.28741965 16.92745638 16.5748393  ... 16.5748393  16.92745638
  17.28741965]
 [17.64003673 17.28007346 16.92745638 ... 16.92745638 17.28007346
  17.64003673]
 [18.         17.64003673 17.28741965 ... 17.28741965 17.64003673
  18.        ]]
```




```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-5, 5, 100)
y = np.linspace(-5, 5, 100)
X, Y = np.meshgrid(x, y)
#Z=X*Y
Z=np.sin(X)*np.cos(Y)
fig=plt.figure(figsize=(14,6))
ax=fig.add_subplot(121,projection='3d')
ax.plot_surface(X,Y,Z,cmap='viridis')
ax2=fig.add_subplot(122)
contour=ax2.contour(X,Y,Z,cmap='viridis')
```

```
fig.colorbar(contour,ax=ax2,shrink=0.5,aspect=5)
plt.tight_layout()
plt.show()
```

